

Dean A. Parker

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Education

New York University, Stern School of Business

Ph.D. in Finance, GPA: 3.71/4

M.Phil. in Finance

Expected May 2027

August 2026

Columbia University

B.S. in Operations Research: Financial Engineering, GPA: 3.84/4

May 2020

Professional Experience

PGIM Quantitative Solutions

June – August 2026

Multi-Asset Research Summer Associate

- Evaluated usage of macroeconomic prediction markets' forecasts and uncertainties as an alpha generating signal for asset allocation.
- Created and backtested strategy negatively correlated with market with information ratio greater than 0.8 after transaction costs with Python.
- Researched ways to forecast commodity returns in response to impact from El Nino events.

AlphaQuest

May 2024 – December 2025

Quantitative Research Intern

- Researched medium-frequency alpha signals in commodity futures and FX markets using statistical modeling and large-scale time-series analysis, contributing to live strategy development and deployment.
- Constructed and evaluated options-based trading signals by applying statistical analysis of FX risk reversal and butterfly spreads to forecast return distributions and skewness.
- Refined machine learning and shrinkage estimators in Python to improve out-of-sample robustness, systematically reducing overfitting across live and backtested regimes using rigorous cross-validation.

Federal Reserve Bank of New York

June 2020 – August 2022

Senior Research Analyst, Capital Markets Function

- Collaborated with economists on monetary policy research published in peer-reviewed academic journals and the Liberty Street Economics blog.
- Rebuilt and enhanced the FRBNY MBS prepayment model in Python and Stata, applying survival analysis and econometric techniques to improve forecast accuracy for internal policy use.
- Performed large-scale data manipulation and visualization in Python and Stata using econometric techniques including local projections and panel regressions to support monetary policy research.

AlphaQuest

November 2019 – June 2020

Quantitative Research Intern

- Built decision tree, gradient boosting, and shrinkage estimators with custom model selection criteria in Python, Cython, and C++.
- Delivered speed and logic improvements to the futures and FX backtesting engine in C#, materially reducing simulation runtime.

Publications

Monetary Policy, Business Liquidity and Survival: Evidence from the Refinancing Channel 2023
With Rodney Ramcharan and Haoyang Liu [Review of Financial Studies](#)

Policy Reports

VC Funds and Regulation D's Rule 506(c): Did Permitting General Solicitation Open the Door for Emerging and Underrepresented Managers? 2024
With Sabrina Howell [SEC](#)

Working Papers

Tyranny of the Personal Network: The Limits of Arm's Length Fundraising in Venture Capital 2025
With Sabrina Howell and Ting Xu R&R Journal of Finance, [SSRN](#)

Mirror, Mirror, on The Wall, Who's the Greenest of Us All? 2025
With Robert Engle [Bloomberg Quant](#), [SSRN](#)

Does Disaster Aid Cause Moral Hazard in the Housing Market? 2025
With Shan Ge Draft In Progress

Leader-Follower Dynamics in Firm Efficiency and Growth 2026
With Robert Engle Draft In Progress

Can Private Be the New Public? Conflicts of Interest in Publicly Listed Private Equity 2026
Dissertation [Draft](#)

Code

STATA Package: twoway_pbar [Github](#)

- Patterned bar charts in Stata.

WRDS Stock Backtesting Engine Work in Progress

- Python code written with Polars to test daily systematic strategies and weighting schemes.

Blog Posts

The Term Spread as a Predictor of Financial Instability 2021
With Moritz Schularick [FRBNY Blog Post](#)

The Housing Boom and the Decline in Mortgage Rates 2021
With Haoyang Liu, David Lucca, and Gabriela Rays-Wahba [FRBNY Blog Post](#), [WSJ](#)

Do People View Housing as a Good Investment and Why? 2021
With Andrew Haughwout, Haoyang Liu, and Xiaohan Zhang [FRBNY Blog Post](#), [Reuters](#)

Languages & Tools

- Programming:** Python (numpy, scipy, polars, pandas, scikit-learn), R (dplyr), Stata, C++, C#, Cython
- Methods:** Panel Regression, Local Projections, Regression Trees, Gradient Boosting, Adaptive Lasso, ML/Shrinkage Estimators, Survival Analysis, Cross-Validation